Xiran Song

Academic Curriculum Vitae

PhD student, Université de Montréal | Mila – Quebec AI Institute
xiran.song@umontreal.ca | https://xiransong.info

Education

Université de Montréal Aug. 2025–present

PhD in Computer Science *(in progress)*; supervisor: Prof. Bang Liu.

Washington University in St. Louis Aug. 2024–July 2025

Doctoral studies in Computer Science *(withdrew from the program)*.

Huazhong University of Science and Technology Sept. 2021–June 2024

Master of Engineering in Computer Science and Technology.

Huazhong University of Science and Technology Sept. 2017–June 2021

Bachelor of Engineering in Computer Science and Technology.

Publications

Conference papers

[1] Zihan Luo, Hong Huang, Jianxun Lian, **Xiran Song**, and Hai Jin. Towards Controllable Hybrid Fairness in Graph Neural Networks. *KDD*, 2025.

[2] **Xiran Song**, Jianxun Lian, Hong Huang, Zihan Luo, Wei Zhou, Xue Lin, Mingqi Wu, Chaozhuo Li, Xing Xie, and Hai Jin. xGCN: An Extreme Graph Convolutional Network for Large-Scale Social Link Prediction. *WWW*, 2023.

[3] Zihan Luo, Hong Huang, Jianxun Lian, **Xiran Song**, Xing Xie, and Hai Jin. Cross-Links Matter for Link Prediction: Rethinking the Debiased GNN from a Data Perspective. *NeurIPS*, 2023.

[4] **Xiran Song**, Jianxun Lian, Hong Huang, Mingqi Wu, Hai Jin, and Xing Xie. Friend Recommendations with Self-Rescaling Graph Neural Networks. *KDD*, 2022.

Journal articles

[5] Zihan Luo, **Xiran Song**, Hong Huang, Jianxun Lian, Chenhao Zhang, Jinqi Jiang, Xing Xie, and Hai Jin. GraphInstruct: Empowering Large Language Models with Graph Understanding and Reasoning Capability. *Frontiers of Computer Science*, 2025 (online publication).

[6] **Xiran Song**, Hong Huang, Jianxun Lian, and Hai Jin. XGCN: A Library for Large-Scale Graph Neural Network Recommendations. *Frontiers of Computer Science*, 2024.

[7] Wei Zhou, Hong Huang, Ruize Shi, **Xiran Song**, Xue Lin, Xiao Wang, and Hai Jin. Temporal Heterogeneous Information Network Embedding via Semantic Evolution. *IEEE Transactions on Knowledge and Data Engineering*, 2023.

Research Experience

Vision-Grounded Motion Generation (VGM) Jan. 2026–present

Université de Montréal / Mila | Supervisor: Prof. Bang Liu

- Formulated VGM to study goal-directed full-body human motion generation from a character's visual observation, recent movement, and a specified target.

- Developed infrastructure for procedural dataset generation, model training, and evaluation. Produced a 13,494-example human motion dataset and conducted controlled studies of motion representation, visual conditioning, and training-data availability.
- Developing RoomTasks-P1, which supports walking and sitting with alternative targets and actions under shared scene observations; full-scale dataset generation is ongoing.

Large Language Models for Graph Understanding Sept. 2023–June 2024

Huazhong University of Science and Technology | Advisor: Prof. Hong Huang

- Investigated the evaluation and improvement of graph understanding and reasoning in large language models; coauthored GraphInstruct.

Large-Scale Graph Representation Learning Sept. 2021–June 2024

Huazhong University of Science and Technology | Advisor: Prof. Hong Huang

- Developed graph neural network models and training strategies for large-scale graph embedding and recommendation, including first-author research on xGCN.

Intern, Microsoft Research Asia Feb.–June 2021

Social Computing Group | Advisor: Dr. Jianxun Lian

- Applied graph neural networks to friend recommendation and developed models aimed at improving Xbox gaming friend-recommendation services, including first-author research on my KDD 2022 paper, "Friend Recommendations with Self-Rescaling Graph Neural Networks."

Scholarships and Distinctions

National Scholarship, Huazhong University of Science and Technology 2022

Merit Student, Huazhong University of Science and Technology 2022

Academic Service

Conference reviewer — KDD Research Track

- KDD 2027, Cycle 1; KDD 2026, Cycle 2; KDD 2025, February cycle.

Journal reviewer

- *IEEE Canadian Journal of Electrical and Computer Engineering*, Aug. 2026.

Research Software

XGCN — Sole developer 2023

A Python library for large-scale graph neural network embedding and recommendation, designed to support million-scale graphs in a single-machine environment.
github.com/CGCL-codes/XGCN_library